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Machine Learning

Exam: IN2064 / Endterm

Date: Monday 19th February, 2024

Examiner: Prof. Stephan Günnemann

Time: 17:00 – 19:00

	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9
I									

Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 38 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____

Problem 1 Decision Tree (4.5 credits)

You want to approach a *regression* problem using a decision tree. You are given the subsequent dataset consisting of a single feature and one regression target.

	Feature x	Target y
1	0.1	0
2	0	-1
3	1.3	4
4	0.4	1

The predictions are obtained via averaging the labels of the corresponding leaf nodes. We construct the tree greedily using binary splits. As split criterion the Squared Error gain $g(\mathcal{Y}, \mathcal{Y}_L, \mathcal{Y}_R) = e(\mathcal{Y}) - e(\mathcal{Y}_L) - e(\mathcal{Y}_R)$ is optimized. $\mathcal{Y}$ denotes the set of regression targets y before the split, while $\mathcal{Y}_L, \mathcal{Y}_R$ are the two sets of regression targets after applying a prospective split. The Squared Error is defined as $e(\mathcal{Y}) = \sum_{y_i \in \mathcal{Y}} (y_i - \bar{y})^2$ with prediction $\bar{y}$.

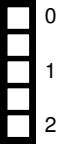
0 ☐ 1 ☐ a) To improve the purity do we need to minimize or maximize the Squared Error gain? **Justify** your answer!

A split becomes perfectly pure if $e(\mathcal{Y}_L)$ and $e(\mathcal{Y}_R)$ both resolve to zero. Thus, we must maximize $g(\mathcal{Y}, \mathcal{Y}_L, \mathcal{Y}_R)$.

0 ☐ 1 ☐ b) Calculate the predictions for all possible binary splits resulting in depth 1 trees.

The possible splits are (a) $\{2\}$ vs. $\{1, 4, 3\}$, (b) $\{2, 1\}$ vs. $\{3, 4\}$, (c) $\{2, 1, 4\}$ vs. $\{3\}$. The predictions are -1 vs. $5/3$, -0.5 vs. 2.5 , and 0 vs. 4 , correspondingly. Of course there are also naïve splits where all samples end up at one leaf node (i.e. $\{\}$ vs. $\{2, 1, 4, 3\}$ and $\{2, 1, 4, 3\}$ vs. $\{\}$). However, these are obviously not improving purity and, therefore, are not considered here.

c) Which of the possible binary splits in (b) results in the best possible split of the data. **Justify** your answer!



For split (a) the Squared error gain resolves to $g(\mathcal{Y}, \mathcal{Y}_L, \mathcal{Y}_R) = e(\mathcal{Y}) - e(\mathcal{Y}_L) - e(\mathcal{Y}_R)$ with $e(\mathcal{Y}) = (-1 - 1)^2 + (0 - 1)^2 + (1 - 1)^2 + (4 - 1)^2 = 14$ and

- (a) $e(\mathcal{Y}_L^{(a)}) = (-1 - (-1))^2 = 0$, $e(\mathcal{Y}_L^{(a)}) = (0 - 5/3)^2 + (1 - 5/3)^2 + (4 - 5/3)^2 = (5/3)^2 + (2/3)^2 + (7/3)^2 = 25/9 + 4/9 + 49/9 = 78/9 = 72/9 + 6/9 = 8 + 2/3$. Thus, $g(\mathcal{Y}, \mathcal{Y}_L^{(a)}, \mathcal{Y}_R^{(a)}) = 14 - 0 - (8 + 2/3) = 5 + 1/3$
- (b) $e(\mathcal{Y}_L^{(b)}) = (-1 - (-0.5))^2 + (0 - (-0.5))^2 = 1/2$, $e(\mathcal{Y}_L^{(b)}) = (1 - 2.5)^2 + (4 - 2.5)^2 = 2.25 + 2.25 = 4.5$. Thus, $g(\mathcal{Y}, \mathcal{Y}_L^{(b)}, \mathcal{Y}_R^{(b)}) = 14 - 0.5 - 4.5 = 9$
- (c) $e(\mathcal{Y}_L^{(c)}) = (-1 - 0)^2 + (0 - 0)^2 + (1 - 0)^2 = 1 + 0 + 1 = 2$, $e(\mathcal{Y}_L^{(c)}) = (4 - 4)^2 = 0$. Thus, $g(\mathcal{Y}, \mathcal{Y}_L^{(c)}, \mathcal{Y}_R^{(c)}) = 14 - 2 - 0 = 12$

Split (c) $\{2, 1, 4\}$ vs. $\{3\}$ is the best since it has the highest purity gain.

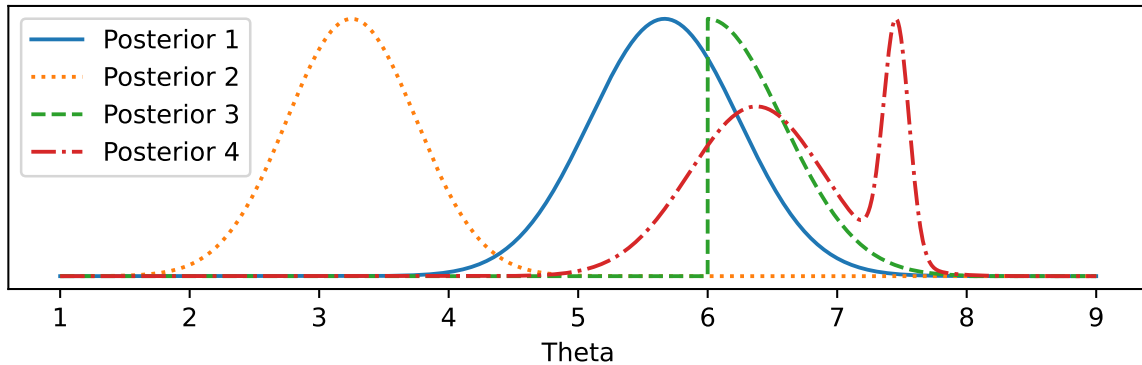
For (a) one could also bound $e(\mathcal{Y}_L^{(a)}) = (0 - 5/3)^2 + (1 - 5/3)^2 + (4 - 5/3)^2 > (1.5)^2 + (1.5)^2 + (2)^2 = 2.25 + 2.25 + 4 = 8.5$ and thus, $g(\mathcal{Y}, \mathcal{Y}_L^{(a)}, \mathcal{Y}_R^{(a)}) < 14 - 0 - 8.5 = 5 + 5$ to come to the same conclusion.

Problem 2 Probabilistic Inference (4 credits)

Consider the probabilistic model

$$p(x | \theta) = \mathcal{N}(x | \theta, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x - \theta)^2\right)$$

and a set of observations $\{x_1, \dots, x_N\} = \{10, 3, 5\}$ consisting of 3 samples $x_i \in \mathbb{R}$. The following figure shows a plot of different unnormalized **posterior distributions** $p(\theta | \mathbf{x})$ over θ . Your task is to match the prior distributions $p(\theta)$ from (a) to (d) to each of the posteriors 1 – 4.



a) $p(\theta) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\theta + 5)^2\right),$

b) Uniform prior, with $a = 6, b = 8,$

c) $p(\theta) = \frac{1}{2} \exp(-|\theta + 5|),$

d) $p(\theta) = 0.5 \frac{1}{0.1\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\theta - 7.5}{0.1}\right)^2\right) + 0.5 \frac{1}{1.2\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\theta - 8}{1.2}\right)^2\right)$

Which posteriors result from applying priors a) to d). **Justify** your answers.

- a) → Posterior 2: This Gaussian prior is unimodal and not truncated, i.e., not 3 and 4. Further, the Gaussian prior penalizes outliers a lot stronger than the Laplace. Thus, the mode of the posterior is comparatively closer to the mean of the prior than to the MLE estimate of 6 (which is also the mode of the Likelihood).
- b) → Posterior 3: Combining a uniform prior with a Gaussian likelihood results in a truncated Gaussian posterior.
- c) → Posterior 1: The Laplace prior is unimodal and not truncated, i.e., it is not 3 and 4. Further, the Laplace prior is less restrictive w.r.t. outliers far away from its mode compared to the unimodal Gaussian with the same location parameter, which results in the mode being closer to the MLE.
- d) → Posterior 4: This mixture of Gaussian priors is multimodal.

Sample Solution

Problem 3 Linear regression (5 credits)

The goal of linear regression is to find an optimal $\mathbf{w}^*$ to approximate $\mathbf{y} \approx \Phi \mathbf{w}^*$ with the design matrix $\Phi \in \mathbb{R}^{n \times d}$.

a) Could we solve the optimization problem by simply computing $\mathbf{w}^* = \Phi^{-1} \mathbf{y}$? **Justify** your answer.

In general $\Phi \in \mathbb{R}^{n \times d}$ is not invertible, e.g., $n \neq d$. For this reason we compute the pseudo inverse instead.

b) The closed form solution for linear regression is given by $\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$. Explain why the pseudo inverse $(\Phi^T \Phi)^{-1} \Phi^T$ might not be computable and provide a possible factor in the data that can lead to it. **Justify** your answer.

While $\Phi^T \Phi$ is positive semi-definite it might be singular, e.g., when columns of Φ are linearly dependent or when we have more features than data samples, which result in $\Phi^T \Phi$ not having full rank.

☐	0
☐	1
☐	2

c) When applying ridge regression the closed-form solution changes to $\mathbf{w}_{ridge}^* = (\Phi^T \Phi + \mu \mathbb{I})^{-1} \Phi^T \mathbf{y}$. **Prove** that if $\mu > 0$ then $(\Phi^T \Phi + \mu \mathbb{I})$ is always invertible.

Hint: A square matrix is invertible iff none of the eigenvalues are zero. Further, any covariance matrix $\Phi^T \Phi$ is positive semi-definite, i.e., for all eigenvalues we know $\lambda \geq 0$.

Let $\mathbf{A} := \Phi^T \Phi$ and $\mathbf{v}$ be an arbitrary eigenvector of $\mathbf{A}$. Then

$$\begin{aligned} (\mathbf{A} + \mu \mathbb{I})\mathbf{v} &= (\mathbf{A}\mathbf{v} + \mu \mathbb{I}\mathbf{v}) \\ &= (\lambda \mathbf{v} + \mu \mathbf{v}) \\ &= (\lambda + \mu)\mathbf{v} \end{aligned}$$

For $\mu > 0$ all eigenvalues of $(\Phi^T \Phi + \mu \mathbb{I})$ are greater than 0 since $\lambda \geq 0$ (remember $\mathbf{A}$ is positive semi-definite). Thus, $(\Phi^T \Phi + \mu \mathbb{I})$ is positive definite and hence, invertible (even if $\lambda = 0$, i.e., $\Phi^T \Phi$ is singular).

Problem 4 Linear Classification (3 credits)

Consider a dataset of scalar datapoints $x_i \in \mathbb{R}$ with binary labels $y_i \in \{0, 1\}$. We now apply a polynomial feature map $\phi_d(x) = (1, x, x^2, \dots, x^d)^\top$ to the datapoints and consider a linear classifier:

$$p(y_i = 1 | x_i, \mathbf{w}) = \sigma(\mathbf{w}^\top \phi_d(x_i)) = \frac{1}{1 + \exp(-\mathbf{w}^\top \phi_d(x_i))}$$

Here, $\sigma(\cdot)$ is the sigmoid function.

Prove that the decision boundary of this classifier is a polynomial of degree d . That is, show that the decision boundary is of the form

$$\sum_{k=0}^d \alpha_k x^k = 0$$

for some $\alpha_k \in \mathbb{R}$ with $k \in \{0, 1, \dots, d\}$.

The decision boundary satisfies $p(y_i = 1 | x_i, \mathbf{w}) = p(y_i = 0 | x_i, \mathbf{w})$.

$$p(y_i = 1 | x_i, \mathbf{w}) = p(y_i = 0 | x_i, \mathbf{w})$$

$$\sigma(-\mathbf{w}^\top \phi_d(x_i)) = 1 - \sigma(-\mathbf{w}^\top \phi_d(x_i))$$

$$\sigma(-\mathbf{w}^\top \phi_d(x_i)) = \sigma(\mathbf{w}^\top \phi_d(x_i))$$

$$-\mathbf{w}^\top \phi_d(x_i) = \mathbf{w}^\top \phi_d(x_i)$$

$$\mathbf{w}^\top \phi_d(x_i) = 0$$

$$\mathbf{w}^\top \phi_d(x_i) = \sum_{k=0}^d \mathbf{w}_{k+1} \phi_d(x_i)_{k+1} = 0$$

$$\sum_{k=0}^d \mathbf{w}_{k+1} x^k = 0$$

Problem 5 Optimization (5 credits)

Consider a convex quadratic function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as:

$$f(x) = a(x - b)^2 + c \quad \text{where } a > 0$$

We want to minimize $f(x)$ using the gradient descent algorithm with a fixed learning rate τ . For which $\tau > 0$ does this method converge to the minimizer b of $f(x)$? **Prove** your answer.

Hint: The method converges if $f(x^{(t+1)}) < f(x^{(t)})$.

We first compute the first derivative of $f(x)$:

$$\frac{\partial f}{\partial x} = 2a(x - b)$$

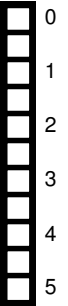
The update step therefore computes $x^{(t+1)} = x^{(t)} - \tau 2a(x^{(t)} - b)$.

We now use the condition in the hint.

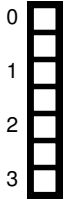
$$\begin{aligned} f(x^{(t+1)}) &= a(x^{(t+1)} - b)^2 + c \\ &= a(x^{(t)} - \tau 2a(x^{(t)} - b) - b)^2 + c \\ &= a((1 - 2\tau a)x^{(t)} - (1 - 2\tau a)b)^2 + c \\ &= a(1 - 2\tau a)^2(x^{(t)} - b)^2 + c \end{aligned}$$

We want $f(x^{(t+1)}) < f(x^{(t)})$:

$$\begin{aligned} a(1 - 2\tau a)^2(x^{(t)} - b)^2 + c &< a(x^{(t)} - b)^2 + c \\ (1 - 2\tau a)^2 &< 1 \\ 2\tau a &< 2 \\ \tau &< 1/a = 2 \left(\frac{\partial^2 f}{\partial x^2} \right)^{-1} \end{aligned}$$



Problem 6 Support Vector Machines and Kernels (3 credits)



Given $a \in \mathbb{R}_{>0}$, $b \in \mathbb{R}$, and $\mathbf{x}, \mathbf{y} \in (0, 1)^n$, i.e., $x_i, y_i \in (0, 1) \forall i \in \{0, \dots, n\}$. **Prove** that

$$k(\mathbf{x}, \mathbf{y}) = \frac{a}{1 - \min_i x_i \min_j y_j} + b^2$$

is a kernel.

Hint: $\frac{a}{1-r} = \sum_{k=0}^{\infty} ar^k$ for $|r| < 1$.

You can use the kernel preserving operations from the lecture, given again as reference.

Let $k_1 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and $k_2 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be kernels, with $\mathcal{X} \subseteq \mathbb{R}^N$. Then the following functions k are kernels as well:

1. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) + k_2(\mathbf{x}_1, \mathbf{x}_2)$
2. $k(\mathbf{x}_1, \mathbf{x}_2) = c \cdot k_1(\mathbf{x}_1, \mathbf{x}_2)$, with $c > 0$
3. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) \cdot k_2(\mathbf{x}_1, \mathbf{x}_2)$
4. $k(\mathbf{x}_1, \mathbf{x}_2) = k_3(\phi(\mathbf{x}_1), \phi(\mathbf{x}_2))$, with the kernel k_3 on $\mathcal{X}' \subseteq \mathbb{R}^M$ and $\phi : \mathcal{X} \rightarrow \mathcal{X}'$
5. $k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^T \mathbf{A} \mathbf{x}_2$, with $\mathbf{A} \in \mathbb{R}^N \times \mathbb{R}^N$ symmetric and positive semidefinite

Using the geometric series, we can rewrite

$$k(\mathbf{x}, \mathbf{y}) = \sum_{k=0}^{\infty} ar^k = \sum_{k=0}^{\infty} \sqrt{a} \sqrt{a} (\min_i x_i)^k (\min_j y_j)^k + b^2 = \phi(\mathbf{x})^T \phi(\mathbf{y}) \quad (6.1)$$

with the feature transformation

$$\phi(\mathbf{x}) = (b, \sqrt{a}, \sqrt{a} \min_i x_i, \sqrt{a} (\min_i x_i)^2, \sqrt{a} (\min_i x_i)^3, \dots) \quad (6.2)$$

Thus, using rule 4, $k(\mathbf{x}, \mathbf{y})$ is a kernel.

Problem 7 Deep Learning (3 credits)

Consider the following code snippet representing the implementation of a function $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, where n is the dimension of the input vectors $\mathbf{x}$ and $\mathbf{y}$. Assume $\mathbf{y} \in \mathbb{R}_{>0}^n$.

```
1 class F:
2     def forward(self, x: np.array, y: np.array) -> np.array:
3         self.cache = x, y
4
5         # MISSING CODE FRAGMENT
6
7         assert out.shape == x.shape
8         assert out.shape == y.shape
9         return out
10
11     def backward(self, d_out: np.array) -> (np.array, np.array):
12         x, y = self.cache
13
14         # CODE FRAGMENT TO BE UPDATED
15         # d_x = d_out.copy()
16         # d_y = d_out.copy()
17         # for i in range(len(x)):
18         #     d_x[i] *= -y[i]
19         #     d_y[i] *= 1 / y[i] - x[i]
20
21         assert d_x.shape == x.shape
22         assert d_y.shape == y.shape
23         return d_x, d_y
```

a) **Update** the highlighted code fragment in the backward method of F (lines 15-19) with the respective vectorized implementation.

```
d_x = d_out * -y
d_y = d_out * (1 / y - x)
```

0
1

b) **Write** one possible vectorized implementation of the forward method of F (missing code fragment on line 5). Multiple lines of code are allowed.

```
out = np.log(y) - x * y + c
where c can be any constant which does not depend on x or y.
```

0
1
2

Problem 8 Autoencoder / Dimensionality Reduction (6.5 credits)

Consider a dataset of real-valued datapoints $\{\mathbf{x}_i\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^D$, i.e., $\mathbf{X} \in \mathbb{R}^{N \times D}$ and $\text{rank}(\mathbf{X}) = r$. We would like to train an autoencoder f with a K -dimensional hidden layer with $K < D$. There are no biases and no activation functions, i.e., $f(\mathbf{x}) = \mathbf{W}_2^T \mathbf{W}_1^T \mathbf{x}$, with $\mathbf{W}_1 \in \mathbb{R}^{D \times K}$ and $\mathbf{W}_2 \in \mathbb{R}^{K \times D}$. f is trained to minimize the following reconstruction error:

$$\mathcal{L} = \sum_{i=1}^N (\mathbf{x}_i - f(\mathbf{x}_i))^T (\mathbf{x}_i - f(\mathbf{x}_i))$$

a) Write down the weights $\mathbf{W}_1$ and $\mathbf{W}_2$ of a global minimizer for the three cases $r > K$, $r = K$, and $r < K$. Justify your answers.

If we write down the autoencoder in matrix notation, we can see that we can apply SVD:

$$f(\mathbf{X}) = \mathbf{X} \mathbf{W}_1 \mathbf{W}_2$$

and the decomposition:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

- $r > K$: we can chose $\mathbf{W}_1 = \mathbf{V}_K$ and $\mathbf{W}_2 = \mathbf{V}_K^T$, where $\mathbf{V}_K$ consists of the eigenvectors corresponding to the K largest eigenvectors.
- $r = K$: we can chose $\mathbf{W}_1 = \mathbf{V}$ and $\mathbf{W}_2 = \mathbf{V}^T$.
- $r < K$: we can chose $\mathbf{W}_1 = \mathbf{V}_0$ and $\mathbf{W}_2 = \mathbf{V}_0^T$, where $\mathbf{V}_0$ consists of the eigenvectors of $\mathbf{V}$ and additional 0 columns.

b) When will the reconstruction loss be zero and how many optimal solutions exist for each of the three previous cases. **Justify** your answers.

0
1

The reconstruction loss will be zero for the two cases $r = K$ and $r < K$. For each solution there exist infinitely many possibilities as we can apply arbitrary rigid transformations to the latent space, i.e., $f(\mathbf{X}) = \mathbf{X}(\mathbf{W}_1 \mathbf{P})(\mathbf{P}^{-1} \mathbf{W}_2)$, where $\mathbf{P} \in \mathbb{R}^{K \times K}$ is a rigid transformation.

c) Assume $\mathbf{W}_1$ is column orthonormal. How do the pairwise Euclidean distances between the original and projected data points change for the cases $r > K$ and $r = K$? **Justify** your answers.

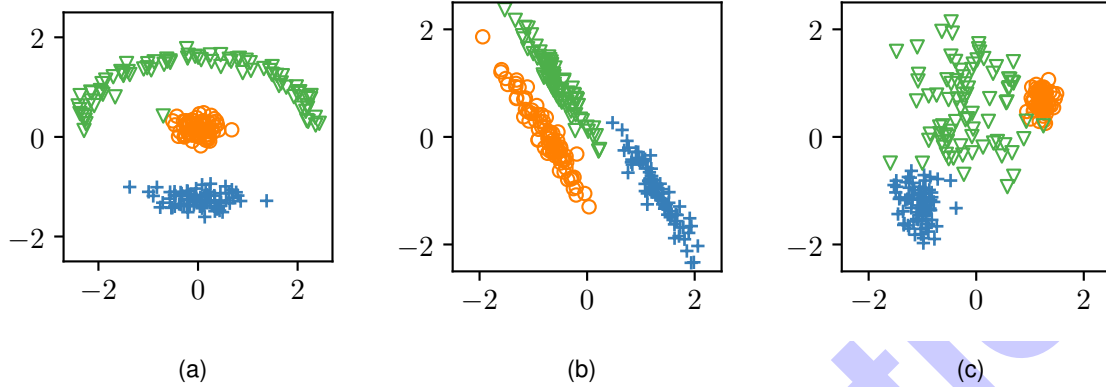
0
1
2

- $r > K$: In general, the distances in the projected space will be smaller than in the data space. As the dimensionality of the latent space is smaller than the inherent dimensionality of the data, the Euclidean distances will decrease. There are special cases where the distance does not change. This happens when they share the same latent space.
- $r = K$: The distances cannot grow, as $\mathbf{W}_1$ can be seen as the first K columns of a rigid transformation. Therefore, the distances can either decrease or be preserved.

Problem 9 Clustering (4 credits)

0 ☐
1 ☐
2 ☐

a) Consider the following three two-dimensional datasets. Each figure shows the most likely cluster assignments based on a Gaussian Mixture Model (GMM) after convergence with $K = 3$ number of clusters. Each symbol represents one cluster assignment.



The table below contains the methods used to create the cluster predictions. Assign each Figure a), b), and c) to their respective clustering method. **Justify** your answers.

	Method	Σ_k
(i)	GMM	$\Sigma_k = \sigma_k I$
(ii)	GMM	No constraints on Σ_k
(iii)	GMM	Σ_k diagonal

- a) The green and blue clusters have different variance along the x and y axis, so the diagonal entries of the covariance matrix can not be equal. However, all clusters can be fit with a diagonal covariance matrix (iii, i.e. axis aligned Gaussians), although the x and y dimension for the green (∇) cluster are clearly correlated. The green (∇) cluster is fitted via a flat Gaussian, and is not a good representation for this cluster.
- b) This can only be solved using the full covariance matrix (ii) with non-zero off-diagonals, because we have non-zero covariance. The variance of the clusters is not axis aligned.
- c) We fit isotropic Gaussians (i), each Gaussian can have different variance. In c) each cluster has different variance and zero covariance, so they can be fitted using an isotropic Gaussian.

b) Show that there exists a degenerate case where the proxy objective used in the expectation maximization (EM) algorithm, and thus the likelihood, becomes unbounded when applying the EM-algorithm to GMMs. **Justify** your answer.

Hint: The probability density function of the multivariate Gaussian with mean $\mu \in \mathbb{R}^D$ and covariance matrix $\Sigma \in \mathbb{R}^{D \times D}$ is

$$\mathcal{N}(\mu, \Sigma) = (2\pi)^{-D/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right).$$

The M-Step for Σ_k is:

$$\begin{aligned} \Sigma_k^{t+1} &= \arg \max_{\Sigma} \mathbb{E}_{\mathbf{Z} \sim \gamma_t(\mathbf{Z})} [\log p(\mathbf{X}, \mathbf{Z} | \pi, \mu, \Sigma)] \\ &= \arg \max_{\Sigma} - \sum_{i=1}^N \sum_{k=1}^K \gamma_t(z_i = k) ((\mathbf{x}_i - \mu_k)^T \Sigma_k^{-1} (\mathbf{x}_i - \mu_k) + \log(\det(\Sigma_k))) \end{aligned}$$

Assume there exists i and k for that $\mathbf{x}_i = \mu_k$ and $\gamma(z_{ik}) \neq 0$. When Σ_k becomes singular, $\det(\Sigma_k) \rightarrow 0$ and the objective is unbounded from above. The Gaussian collapses to one point. The other mixture components can still have finite variance. There is nothing that would balance this in comparison of fitting a single Gaussian.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper, consisting of 30 columns and 40 rows of small squares. A diagonal watermark with the text "Sample Solution" in a light blue, sans-serif font is overlaid across the grid, running from the bottom-left towards the top-right.